



Reclamation of beneficial bioactivities of herbal antioxidant condensed tannin extracted from *Euonymus laxiflorus*

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Abstract

Natural condensed tannins (CT) have received great interest due to their vast beneficial bioactivities. Recently, *Euonymus laxiflorus*, an edible medicinal herb collected in the Central Highlands of Vietnam, was newly recognized as a natural source rich in antioxidant CT. The aim of this study is to scale up extraction and investigate various novel bioactivities of the isolated CT. The screening of the parts used rich in CT, such as heartwood, trunk bark, branch, and the leaves of EL, indicated that the EL trunk bark (ELTB) has the highest CT content. *E. laxiflorus* CT (ELCT) was extracted with high-level scale from ELTB via several techniques and then was qualified by NMR spectra and HPLC finger printing. A condensed tannin ELBT-3.1.d was isolated and showed its high molecular weight of ≥ 10 kDa and various in vitro bioactivities. ELBT-3.1.d demonstrated high antioxidant capacity ($IC_{50}=6.11 \mu\text{g}/\text{mL}$), and potent enzymes inhibition targets in anti-diabetes, especially showing novel potential inhibition against human salivary α -amylase with IC_{50} value of $3.01 \mu\text{g}/\text{mL}$. ELBT-3.1.d displayed no anti-NO effect but did show moderate to effective anticancer activities against A549, MCF-7, HepG2, and WiDr. Notably, condensed tannin ELBT-3.1.d displayed no toxicity to normal cells and a novel potential prebiotic effect on *Lactobacillus paracasei* BCRC14023 and *Lactobacillus rhamnosus* BCRC16000. In the animal test, ELCT demonstrated significant effect on reducing plasma glucose in mice at the doses of 50–100 mg/kg BW. The current study contributes to enrich the novel bioactivities of ELCT which may be suggested as functional food or drugs due to processing numerous beneficial bioactivities.

Keywords *Euonymus laxiflorus* · Antioxidants · Anti-NO · Prebiotic activity · Condensed tannins · Hypoglycemia

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Introduction

Condensed tannins or proanthocyanidins are a class of polyphenols. They are polymers of flavans containing no sugar moieties and were early recognized as bioactive molecules, showing numerous bio-functions, including antioxidant activity [1], enzymes inhibition and hypoglycemic effect [2, 3], gastrointestinal nematodes inhibition [4], potentially inhibited animal fatty acid synthase, and breast cancer cells [5]; therefore, condensed tannins were suggested as a valuable resource of bioactive substances holding vast beneficial effects. As such, these molecules have received much attention in investigations of extraction and detection of novel biological activities from various herbal resources [3, 5].

Euonymus laxiflorus (EL) is an edible medicinal herb growing wildly in the Central Highlands of Vietnam [3, 6] and also commonly distributed in China, India, Myanmar, and Cambodia [7]. In recent years, *E. laxiflorus* (EL) has been investigated for its novel biological activities, including antioxidant, anti-NO, enzymes inhibitory, hypoglycemic, and anti-diabetic activities [6, 8–10]. This herbal species has also been found to be a rich potential source of bioactive compounds [2, 3, 8, 11, 12]. Various compounds identified from the leaves and the trunk bark of this herb have been reported to show potent antioxidant and anti-nitric oxide effects [8, 11, 12]. Novel α -amylase inhibitors [2] and hypoglycemic constituents [3], antioxidant, and anti-NO compounds [8] were isolated and identified from the trunk bark of EL. In our previous study, an antioxidant condensed tannin (CT) was isolated from the trunk bark of *E. laxiflorus* and reported for its enzymic inhibition and hypoglycemic effects [2, 3]. However, there has been no report on the extraction of CT from this potential medicinal plant so far to secrete the part rich in CT and some basic condition effects on the extraction of *E. laxiflorus* condensed tannin with high scale. Very few biological activities of *E. laxiflorus* condensed tannin (ELCT) were reported beyond the above cited studies [2, 3], especially regarding the lack of safety information of ELCT reported to date.

As a part of the studies investigating and developing the major components of *E. laxiflorus* extract as health food or drugs, this study focused on enhancing CT extraction via screening the parts used, suitable solvent, temperature extraction, and investigation of various valuable bio-functions, including antioxidant, inflammatory, human salivary α -amylase inhibitory, and prebiotic activities, the effect on reducing plasma glucose in mice, as well as reporting on the toxicity of this molecule on normal and cancerous cells.

Materials and methods

Materials

The medicinal plant parts used in the samples (*E. laxiflorus*) included heartwood, trunk bark, branch, and leaves collected from Yok Don National Park, Dak Lak

Province, Vietnam. 2-diphenyl-1-picrylhydrazyl, *S. cerevisiae* α -glucosidase and acarbose were purchased from Sigma Chemical Co. (St. Louis City, MO, USA). *p*-nitrophenyl glucopyranoside (*p*NPG) was obtained from Sigma-Aldrich, St. Louis, MO, USA. Porcine pancreatic (type VI-B) and human saliva (type IX-A) α -amylases were obtained from Sigma-Aldrich, USA. Diaion HP-20, Rat intestinal acetone powders, and octadecylsilane (ODS) gel were purchased from Mitsubishi Chemical Co. (Tokyo, Japan), Sigma-Aldrich (Singapore), and Merck, respectively.

Extraction and content determination of condensed tannin from *E. laxiflorus*

All the parts used, including heartwood, trunk bark, branch, and the leaves of *E. laxiflorus*, were powdered and then extracted by using the method described in detail in a previous study [6]. The medicinal plant parts used (5 g) were extracted with methanol (50 mL) in a glass conical flask using a shaker (150 rpm) for 24 h at 60 °C and then filtrated with the filter paper (No. 1; Whatman International, Maidstone, UK). The residue was extracted twice with the same above method. All 3 methanol extracts were mixed and then evaporated in a rotary evaporator (IKA, Germany) at 60 °C. The trunk bark was found to possess the highest condensed tannin content. Thus, it was further extracted by various solvents (methanol, ethanol, ethyl acetate, butanol, and water) by the same method described above. Methanol and water showed the most suitability for condensed tannins extraction, and their mixtures at different ratios (methanol in water: 0, 25 50, 75, and 100%) were further investigated; 75% methanol was found to be the best, and as such was used to investigate the effect of temperature extraction (30, 40, 50, 60, 70 °C) on condensed tannin content, and temperature extraction at 50 °C was used for condensed tannin extraction with the highest yield. The 75% methanol extract of *E. laxiflorus* trunk bark was used for isolating the condensed tannin according to the method presented in the previous report [2], and the process of isolation is summarized in Fig. 1. The sub-fraction ELTB.3.1-2d showing the highest content of CT was isolated and qualified and then used for the detection of biological activities. The content of condensed tannin was determined according to the butanol-HCl method [13] qualified by NMR spectra and HPLC finger printing following the method reported in our previous study [2].

In vitro biological activity assays

Antioxidant capacity was evaluated by using the DPPH radical scavenging activity method described by Nguyen et al., 2018 [14] with minor modification. Eighty microliters of samples was mixed with 20 μ L of 0.75 mM DPPH in methanol in a plate with 96 wells kept in dark for 30 min. The optical density was measured at the wavelength at 517 nm. The antioxidant activity was calculated using the following formula: Activity (%) = $\left(\frac{A-B}{A}\right) \times 100\%$, where *A* is the optical density of the blank sample (no antioxidants in the mixture solution) and *B* refers to the optical density of the sample solution after 30 min of reaction. The

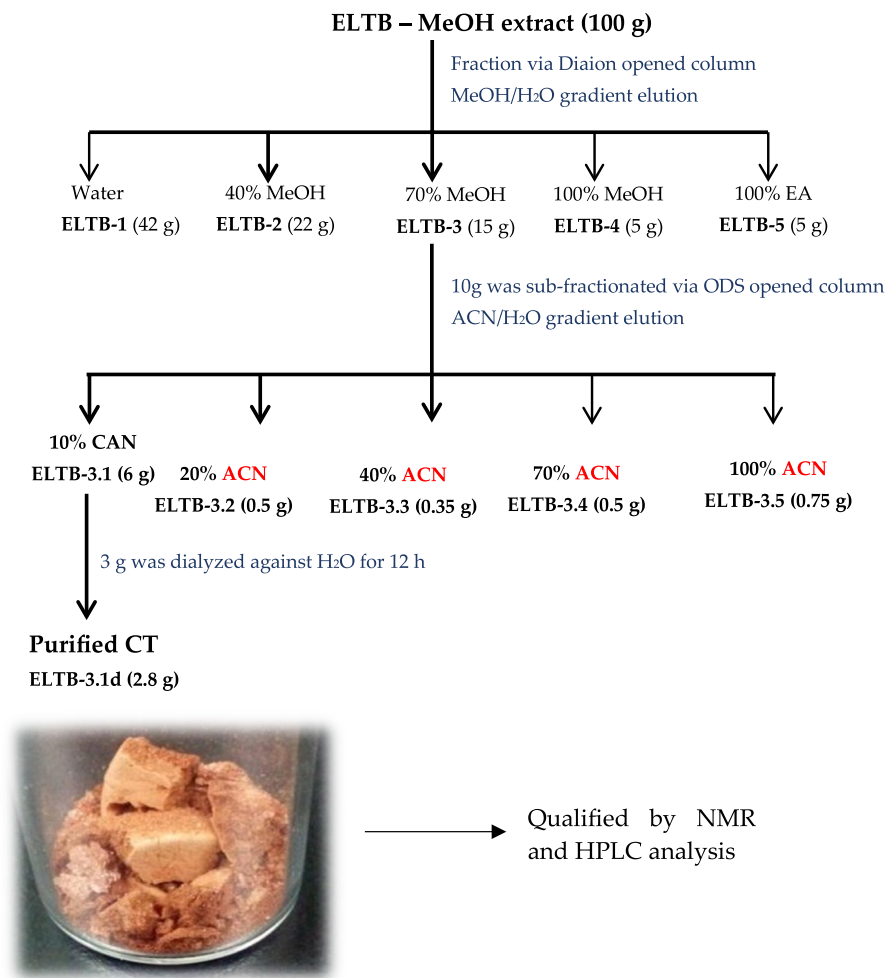


Fig. 1 Isolation chart of condensed tannins (CT) from MeOH extract of *Euonymus laxiflorus* trunk bark (ELTB) extract. ODS: octadecylsilane; ACN: acetonitrile

activity was also expressed as the IC₅₀ value, defined as the concentration of an antioxidant which may reduce 50% the density of the purple color of the DPPH solution. α -Tocopherol was used as a positive control for the comparison purpose.

Anti-NO activity, anticancer effect, and prebiotic property were performed according to the methods in detail described by Nguyen et al. [14], Nguyen et al. [15], and Doan et al. [16], respectively. The enzyme inhibitory activity against α -glucosidases and α -amylases was done following the assays mentioned in detail by Nguyen et al. [17], and Nguyen et al. [10], respectively.

Animal tests

The reducing plasma glucose in mice of condensed tannin ELTB.3.1-2d was conducted according to the assay mentioned in an earlier report [17]. Thirty-two male ICR (Institute of Cancer Research) mice (at 7 weeks old) supplied from The National Laboratory Animal Center (No. 128, Sect. 2, Academia Rd., Nangang Dist., Taipei City, Taiwan) were divided into 4 groups based on equivalent mean plasma glucose levels and body weight, with 8 mice per group. One control group were orally administered with distilled water. Three other groups were administered 50 and 100 mg/kg BW of condensed tannin ELTB.3.1-2d and 50 mg/kg BW of acarbose. These tests were carefully performed according to the guidelines and approval of the IACUC (Institutional Animal Care and Use Committee of the National Research Institute) of Chinese Medicine, Ministry of Health and Welfare (IACUC No.104-706-1, 29/12/2014).

All the mice fasted overnight for 16 h and were administered distilled water, condensed tannin ELTB.3.1-2d or acarbose; after 20 min, the mice were loaded with a sucrose solution at 3 g/kg BW. Plasma blood was sampled and measured after 0, 0.5, 1.0, 1.5, and 2.0 h.

Results and discussion

Enhancing extraction of condensed tannins from *E. laxiflorus*

To enhance the condensed tannins extraction from *E. laxiflorus*, some major factors were tested, such as parts used, solvent, and temperature-related extraction. For the first extraction of natural products, methanol was often used [8, 18, 19]; thus, this solvent was employed to extract all the parts used, including heartwood, trunk bark, branch, and the leaves, using the method described in detail in the method section. The result illustrated in Fig. 2a indicated that the EL trunk bark (ELTB) has the highest CT content of 421 mg/g, followed by the leaves, heartwood, and branch with the CT contents of 247, 165, and 152 mg/g, respectively. Therefore, ELTB was chosen as the richest source of CT for further investigation. Among various solvents used for extraction, methanol demonstrated the best suitability for harvesting the highest CT yield of 424 mg/g (Fig. 2b), reaching extraction max yield of 527 mg/g when 75% methanol in water was used as the extraction solvent (Fig. 2c). The temperature extraction also showed significant effect on CT yield extraction. The CT yield reached the highest level at the extraction temperature at ≥ 50 °C (Fig. 2d). To save energy in industrial CT extraction with fewer lost bioactivities of CT, the extraction temperature at 50 °C was the best choice for the extraction of CT from ELTB.

The purification method of CT from ELTB was conducted via several techniques, including a Diaion opened column, an ODS opened column, and dialysis according to the previous report [2]. The purification process is summarized in Fig. 1. The crude extract rich in CT was primarily separated into 5 fractions via a Diaion opened column. Fraction 3 (ELTB-3, eluted with 70% MeOH) was found to have the highest

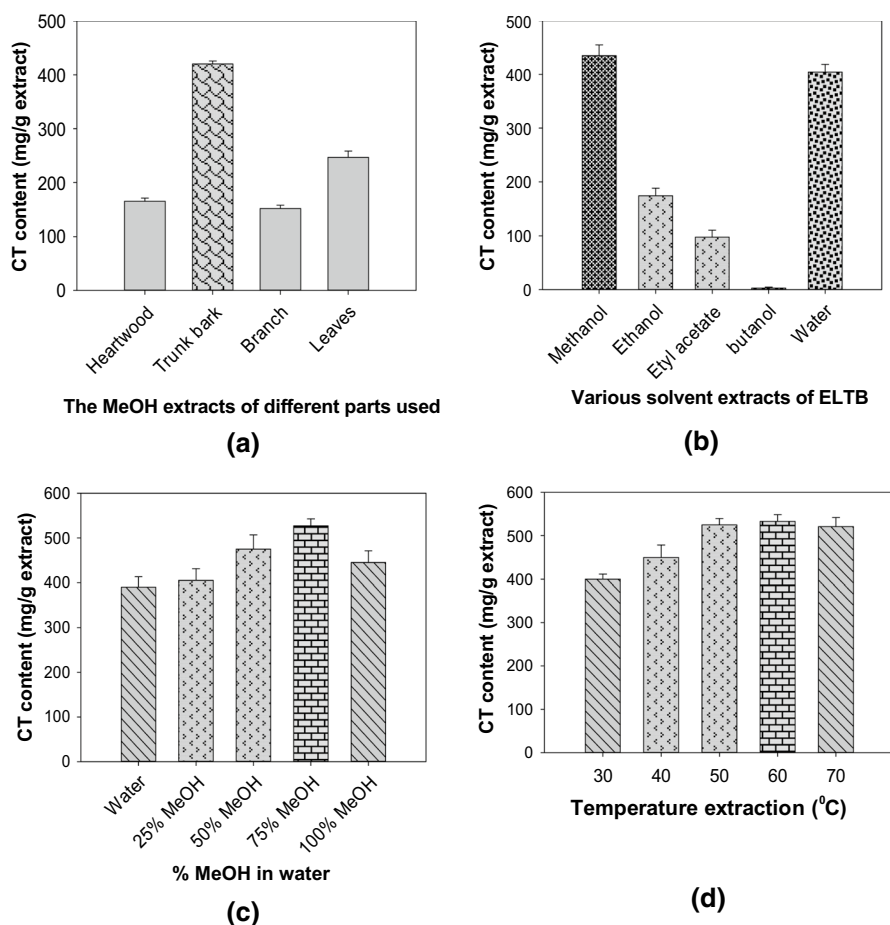


Fig. 2 Screening the parts used rich in condensed tannin (CT) and investigating some factors affecting CT extraction. Different parts used of *Euonymus laxiflorus* were extracted by MeOH at 60 °C (a); the *Euonymus laxiflorus* trunk bark (ELTB) was extracted with various solvents at 60 °C (b); the ELTB was extracted with the mixture of MeOH and water at different ratios at 60 °C (c); the ELTB was extracted with 75% MeOH in water at different temperatures (d)

CT content (790 mg/g); as such, this fraction was further sub-fractionated by an ODS opened column, and the sub-fractions eluted with 10% ACN was screened as the highest CT content (94.2 mg/g). This sub-fraction was further purified by dialysis to harvest the substrate (ELTB-3.1d) with the most CT content (989.7 mg/g). The ^{13}C - NMR and UV spectra were used to detect and qualify the CT during the isolation process. In the previous study, the CT showed the ^{13}C - NMR data (150 MHz, MeOH- d_4 , δC ppm: 161.0, 155.0, 144.5, 138.5, 132.0, 122.0–118.0, 116.3–114.0, 107.0, 102.0, 97.0, 76.0, 35.5, and 34.0) [2]. As shown in Fig. 3a–d, the signals of ^{13}C - NMR data clearly appeared after the sample was separated via a Diaion opened column, and the C signals of this CT were clearer after the purification steps.

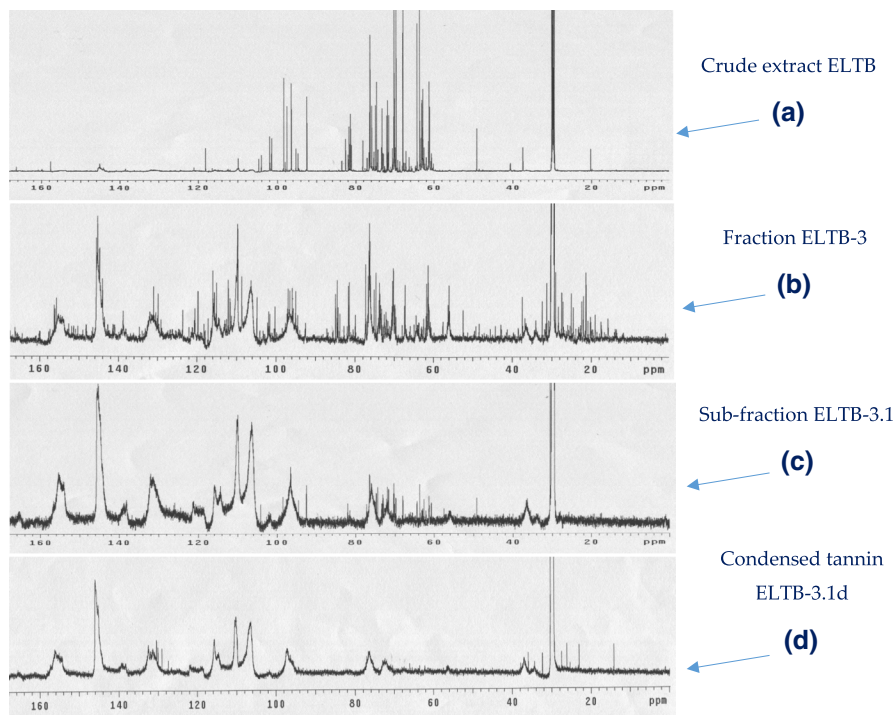


Fig. 3 ^{13}C NMR spectra of crude extract ELTB (a), fraction ELTB-3 after Diaion gel separation (b), sub-fraction ELTB-3.1 after ODS gel sub-separation (c), and condensed tannin ELTB-3.1d obtained after dialysis (d). ^{13}C NMR was measured in MeOH-d_4 at 150 MHz [2]

The ^{13}C - NMR data of this CT completely fit with those of the CT isolated in the previous report [2] and were closely related to those of *Delonix regia* CT [20] and persimmon tannin [21].

The condensed tannin was also detected by HPLC analysis. As shown in Fig. 4a–d, the condensed tannin achieved high purity after just 2 steps of purification (Diaion opened column and ODS opened column). The peak of this compound expanded. In addition, the CT were obtained from the dialysis method using the dialysis tube with the size of 10 kDa, and the CT were collected inside; thus, the evidence indicated that *E. laxiflorus* CT should be macromolecular with high molecular weight (≥ 10 kDa) [2], the same as the reported CT. The persimmon tannin also showed high molecular weights of 5.574 kDa (gel permeation chromatography method) and 10.749 kDa (thiolysis MW method) [2].

Evaluation of biological activities of condensed tannins isolated from *E. laxiflorus*

With the aim of reclamation of CT extracted from *E. laxiflorus* as the rich source of bioactivities, various beneficial effects were tested, including antioxidant,

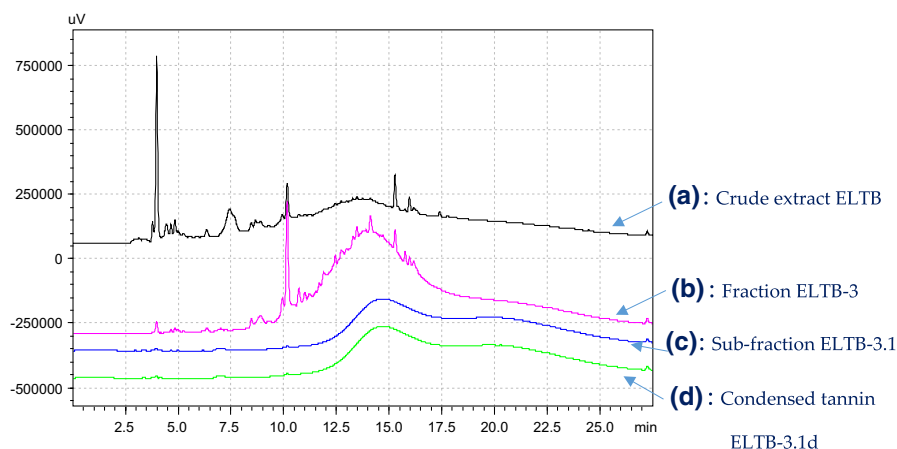


Fig. 4 HPLC fingerprints of crude extract ELTB (a), fraction ELTB-2 after Diaion gel separation (b), sub-fraction ELTB-3.1 after ODS gel sub-separation (c), and condensed tannin ELTB-3.1d obtained after dialysis (d). HPLC was analyzed using a C18 column at 25 °C with a flow rate of 0.8 mL per min, mobile phase condition: 0–5' (5–30% ACN), 5–15' (30–70% ACN), 15–25' (70–100% ACN); 25–30' (100% ACN) and ultraviolet (UV) detected at 245 nm. Acetonitrile (ACN) [2]

inflammatory, enzyme inhibitory, and prebiotic activities, glycemia in mice, and toxicity on normal and cancerous cells.

Antioxidant, anti-NO, and anticancer activities of condensed tannin ELTB.3.1-2d

Free radicals are believed to cause vital damage to lipids, proteins, and the DNA of living organisms; as such, they may cause array of diseases [22]. These harmful effects may be reduced with the use of antioxidants. Thus, antioxidants produced by chemical synthesis were used for protection against such damage and reported causing several inverted effects [23]. Therefore, seeking abundant and natural resources of antioxidant compounds is of interest and continual needed. In this study, DPPH radical scavenging capacity was used to evaluate the antioxidant efficacy of the samples. The crude extract rich in CT and purified CT was used for the detection of DPPH radical scavenging activity. The activity of α -Tocopherol, a commercial antioxidant compound, was also tested for comparison purpose. The data shown in Fig. 5 indicate that CT purified from *E. laxiflorus* (ELBT-3.1.d) demonstrated high antioxidant capacity with great max activity (100%) at low concentration of 25 μ g/mL, while the crude sample (ELBT) and α -Tocopherol reached max activity of 94.5% and 96.5%, respectively, at higher concentration of 100 μ g/mL. For clearer results of the evaluation of antioxidant capacity, the activity was also expressed as IC₅₀ values (Table 1), defined as the concentration of antioxidant which may reduce 50% of radical scavenging. Thus, the lowest IC₅₀ value the antioxidant gets, the strongest activity it possesses. As shown in Table 1, CT displayed the highest activity due to its smallest IC₅₀ values of 6.11 μ g/mL.

Recently, various antioxidant compounds were also isolated and identified from the same medicinal plant *E. laxiflorus* [8]. In the comparison summarized

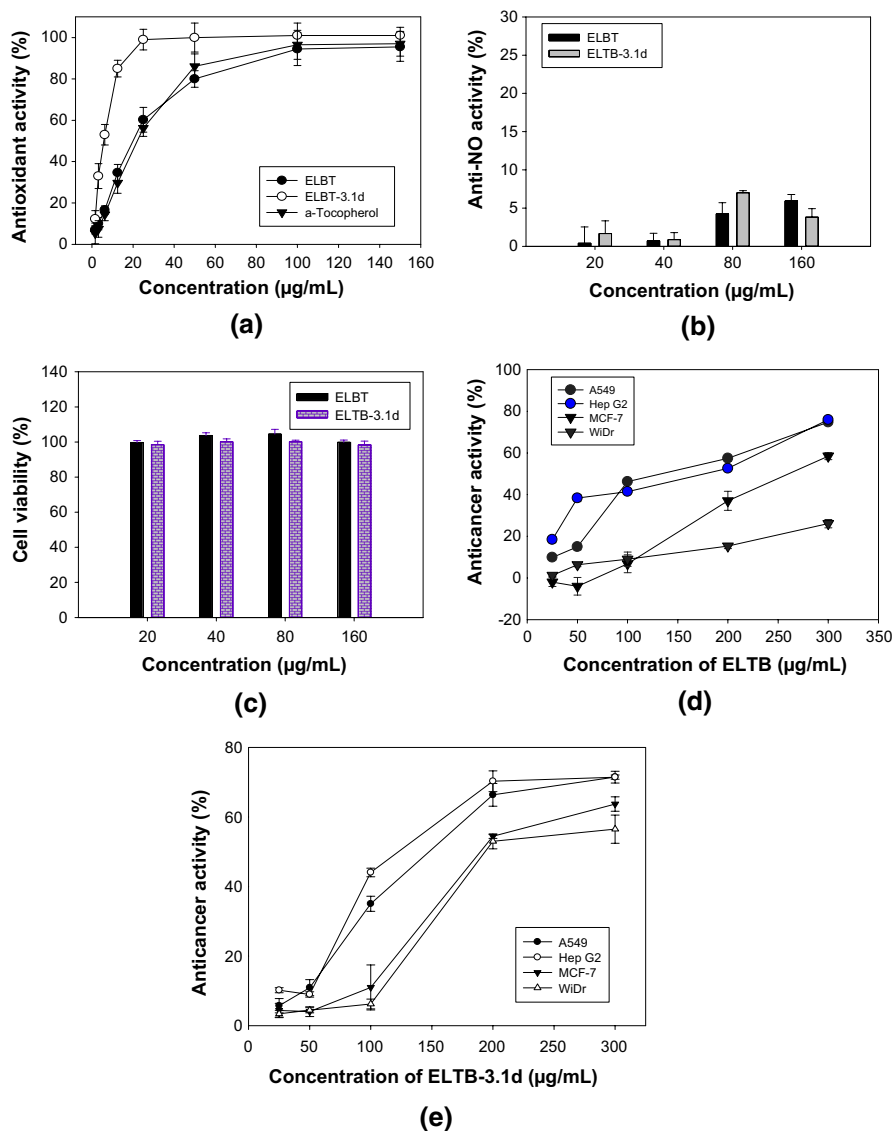


Fig. 5 DPPH radical scavenging capacity (a), anti-NO activity (b), effect on cell viability (c) of crude extract ELTB, and condensed tannin ELTB-3.1d. Anticancer activity against some cancerous cell lines: A549 (human lung carcinoma), MCF-7 (human breast adenocarcinoma), Hep G2 (human hepatocellular carcinoma), and WiDr (human colon adenocarcinoma) of crude extract ELTB (d), and condensed tannin ELTB-3.1d (e)

in Table 1, the CT compound purified in this study was one of the most potential antioxidants since it showed the smallest IC₅₀ value (6.11 $\mu\text{g/mL}$) compared to other reported antioxidant compounds (7.1–58.11 $\mu\text{g/mL}$). The evidence in this study confirmed that genus *E. laxiflorus* is a rich source of bioactive antioxidant

Table 1 Evaluation of the antioxidant activity of the isolated compounds

No.	Isolated compounds	IC ₅₀ (μg/mL)	References
1	Condensed tannin ELTB-3.1d	6.11	This study
2	Crude extract ELTB	20.2	This study
3	α-Tocopherol	22.5	This study
4	2-methoxy-4-hydroxyphenol-1- <i>O</i> -β-D-glucopyranoside	38.09	[8]
5	1-β-D-glucopyranosyloxy-3,5-dimethoxy-4-hydroxybenzene	58.11	[8]
6	3,4-Dihydroxybenzoic acid	16.77	[8]
7	Walterolactone A/B 6- <i>O</i> -gallate-β-D-glucopyranoside	10.9	[8]
8	Walterolactone A (1a) and B (1b) β-D-glucopyranoside	28	[8]
9	Gallic acid	9	[8]
10	(-)-Gallic acid	30.73	[8]
11	Leonuriside	27.47	[8]
12	(+)-Catechin	7.10	[8]
13	Methyl galloate	9.4	[8]
14	(-)-Catechin	11.5	[8]
15	(3,5-dimethoxy-4-hydroxyphenol)-1- <i>O</i> -β-D-(6'- <i>O</i> -galloyl)-glucopyranoside	7.8	[8]

compounds and CT may be suggested as a potential candidate of antioxidant for alternative use.

The pro-inflammatory activity (Fig. 5b) and the effect on normal cells (Fig. 5c) of the crude extract ELTB and condensed tannin ELTB-3.1d were evaluated by using LPS-stimulated RAW 264.7 cells [8]. As shown in Fig. 5b, both ELTB and ELTB-3.1d exhibited weak anti-NO activity ($\leq 7\%$) even at their high concentration of 160 μg/mL and showed no toxicity at all to the normal cells (Fig. 5c) at the same tested concentration. The anticancer activities were also tested against some cancerous cell lines models, including A549 (human lung carcinoma), MCF-7 (human breast adenocarcinoma), Hep G2 (human hepatocellular carcinoma), and WiDr (human colon adenocarcinoma). As shown in Fig. 5d, the crude extract ELTB showed effective anticancer activity against A549 and HepG2, moderate inhibition against MCF-7, and weak activity against WiDr with the inhibition values of 74.8%, 76.1%, 58.5%, and 26.07%, respectively, at its concentration of 300 μg/mL. Condensed tannin extracted from *E. laxiflorus* also demonstrated effective inhibition against A549 and HepG2 cancerous cell lines (66.4–70.3% at its concentration of 200 μg/mL) and moderate activity against both MCF-7 and WiDr (53.0 – 54.44% at its concentration of 200 μg/mL) (Fig. 5e). Condensed tannins, such as (-)-epigallocatechin-3-gallate (EGCG), gallotannins, and ellagitannins, were widely reported as showing anticancer properties [24]. However, there are little available data on the anticancer activity of condensed tannins with high molecular weight. In this study, the condensed tannin ELTB-3.1d with molecular weight ≥ 10 kDa isolated from *E. laxiflorus* trunk bark was reported with novel promising anticancer effects against some tested cancerous cell lines: A549, HepG2, MCF-7, and WiDr.

Enzymatic inhibition targets in type 2 diabetes and the hypoglycemic effect in mice of condensed tannin ELTB.3.1-2d

Phenolic constituents have been reported to exhibit vast beneficial effects, including anti-diabetic and hypoglycemic effect [1–3, 8, 24–29]. To investigate the potential use of condensed tannin ELTB.3.1-2d, this constituent was evaluated for its inhibition against some enzyme targets in drugs for type 2 diabetes, including *S. cerevisiae* α -glucosidase, rat intestinal α -glucosidase, porcine pancreatic α -amylase, and human salivary α -amylase. The results were expressed as IC₅₀ values and are summarized in Table 2. Condensed tannin ELTB.3.1-2d displayed high inhibition against *S. cerevisiae* α -glucosidase, porcine pancreatic α -amylase, and human salivary α -amylase with very low IC₅₀ values of 1.04, 1.08, and 3.01 μ g/mL, respectively, while demonstrating moderate activity against α -glucosidase from rat with the IC₅₀ value of 500 μ g/mL. The crude extract (ELTB) and acarbose were also tested for comparison purpose. ELTB also strongly inhibits *S. cerevisiae* α -glucosidase, porcine pancreatic α -amylase, and human salivary α -amylase with IC₅₀ values in the range of 1.33–7.3 μ g/mL; however, ELTB displayed very weak inhibition against rat intestinal α -glucosidase with high IC₅₀ value of 1986 μ g/mL. Acarbose, a commercial anti-diabetic drug, showed good effect against rat intestinal α -glucosidase, porcine pancreatic α -amylase, and human salivary α -amylase, but very weak inhibition against *S. cerevisiae* α -glucosidase with IC₅₀ values of 125, 154, 103, and 1120 μ g/mL. Recently, yeast *S. cerevisiae* α -glucosidase and porcine pancreatic α -amylase have been widely used as indicators of anti-diabetic drugs in various publications. However, rat α -glucosidase and human salivary α -amylase may be better indicators of anti-diabetic drugs as human salivary α -amylase originated from human and rat α -glucosidase (animal enzyme) is closer to that of humans [6]. In this study, ELTB-3.1d showed moderate inhibition against α -glucosidase from rat and potential effect on human salivary α -amylase. Thus, ELTB-3.1d may be a good candidate for type 2 diabetes management agent.

For further indications of the potential effects of ELTB-3.1d in animals, this constituent was tested for its effect on reducing plasma glucose in mice at the 2 concentration of 50 and 100 mg/kg BW. As shown in Fig. 6, ELTB-3.1d displayed significant effect on reducing plasma glucose in mice at 50 mg/kg BW at 0.5 to 1.0 h after the mice were loaded with sucrose. At higher dose of 100 mg/kg BW, ELTB-3.1d demonstrated more significant active on reducing plasma glucose until 2 h after the

Table 2 Enzymatic inhibition targets in type 2 diabetes

Enzymes	Enzyme inhibition of samples (IC ₅₀ μ g/mL)		
	ELTB	ELTB-3.1d	Acarbose.
<i>S. cerevisiae</i> α -glucosidase	1.33	1.04	1120
Rat intestinal α -glucosidase	1986	500	125
Porcine pancreatic α -amylase	7.3	1.08	154
Human salivary α -amylase	3.5	3.01	103

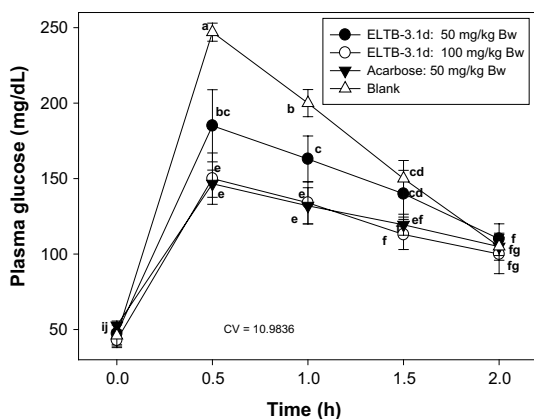


Fig. 6 Hyperglycemic effect of condensed tannin ELTB.3.1-2d and acarbose in a mouse model. Institute of Cancer Research (ICR) mice were used as the animal model. ELTB.3.1-2d at the doses of 50 and 100 mg/kg BW and acarbose at a dose of 50 mg/kg BW were administered to the ICR mice. In the blank group, distilled water was administered to mice, and the mice were loaded with a sucrose solution at 3 g/kg BW after 20 min. The mice blood was sampled and measured at 0.5, 1.0, 1.5, and 2.0 h after sucrose was loaded into the mice. Different letters of means of plasma glucose level are significantly different ($\alpha=0.05$) based on Duncan's multiple range test. CV (coefficient of variation) = 10.9836

sugar loaded to mice. The result also clearly indicated that ELTB-3.1d at 100 mg/kg BW showed its comparable activity to acarbose, a commercial anti-diabetic drug. At the dose of 50 mg/kg BW, acarbose demonstrated a strong effect; however, some mice (3/8 mice) showed symptoms of diarrhea while no mice in the groups treated with ELTB-3.1d at 50 and mg/kg BW showed such symptoms.

Natural products like coffee and tea, rich in polyphenolics content, were investigated for their anti-hyperglycemic effect in mice and reducing effect on post-prandial plasma glucose in humans [30, 31]. Several extracts of medicinal plants rich in content of condensed tannins were also reported showing anti-hyperglycemic effects in rats [9, 32, 33]. The crude extract of pinhão coat and tannin extracted from the *A. mearnsii* were found to significantly reduce the post-prandial glycemic levels in rats at the doses of 250 mg/kg bw. The extract of *Psidium littorale* Raddi leaf showed an effective anti-hyperglycemic effect in streptozotocin-induced diabetic rats at the dose of 150 mg/kgBW [9]. Recently, *E. laxiflorus* trunk bark extract was tested for its effect in diabetic rats at a dose of 200 mg/kg BW; hyperglycemia was significantly reduced to the same level of acarbose at 120 mg/kg BW [32]. In this study, condensed tannin ELTB.3.1-2d demonstrated significant effect at the low doses of 50-100 mg/kg BW via using a sucrose tolerance test. In our previous report, *E. laxiflorus* condensed tannin also showed its potency in reducing glycemia at the dose of 100 mg/kg BW via using starch tolerance test [3]. It was proved that starch is degraded by the action of amylases to smaller molecules, dextrin oligomers which are further converted into α -D-glucoses by catalysis of α -glucosidase. These α -D-glucoses may be absorbed into the blood circulation intestinally, leading to hyperglycemia. Thus, the use of enzyme inhibitors of α -amylase and α -glucosidase may slow down or block this process, resulting in anti-hypoglycemic activity [3].

Condensed tannin extracted from *E. laxiflorus* possessed both potent α -amylase and α -glucosidase inhibitory activities (Table 2). Therefore, it demonstrated potent hypoglycemic effects in mice when using starch or sucrose tolerance tests.

The results provided in this study and in our previous reports [3, 9] indicated that *E. laxiflorus*, a natural source rich in condensed tannin, may have potential for functional foods or anti-diabetic and hypoglycemic drugs.

The effect of condensed tannin ELTB.3.1-2d on the growth of lactic bacterial strains

The detection of the effect of condensed tannin ELTB.3.1-2d on the growth of lactic bacterial strains, including *Lactobacillus paracasei* BCRC14023 and *Lactobacillus rhamnosus* BCRC16000, was tested. These bacteria were cultured in MRS broth (control) and MRS broth supplemented with condensed tannin ELTB.3.1-2d at a concentration of 156 $\mu\text{g/mL}$, and the bacterial growth was detected during 24 h of fermentation. As shown in Fig. 7, condensed tannin ELTB.3.1-2d enhanced the

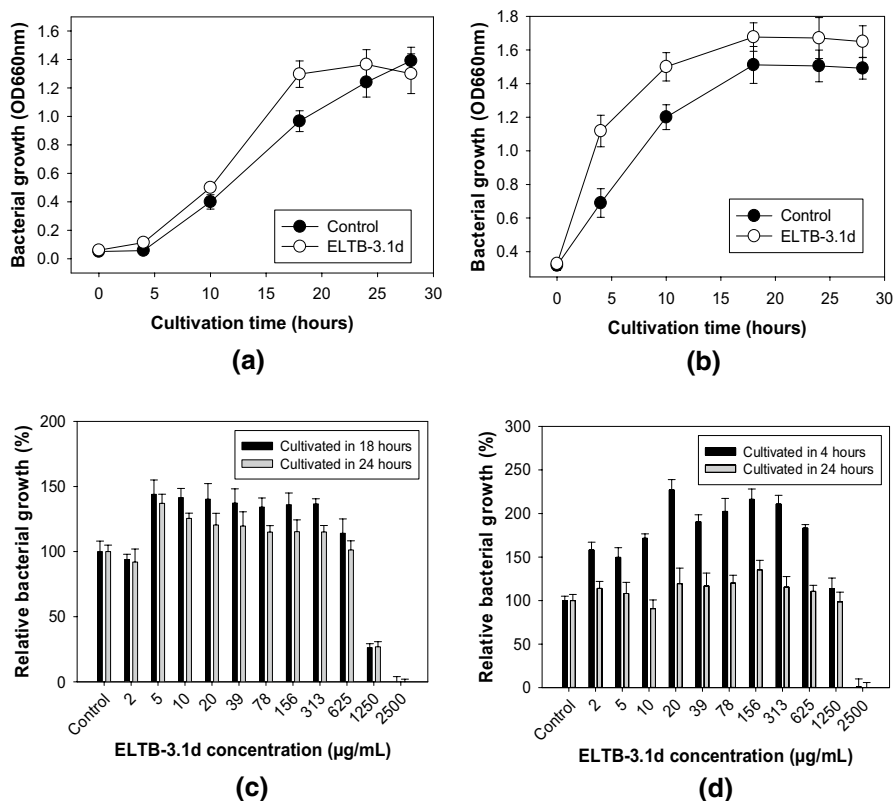


Fig. 7 Effect of condensed tannin ELTB.3.1-2d on the growth of *Lactobacillus paracasei* BCRC14023 (a, c) and *Lactobacillus rhamnosus* BCRC16000 (b, d) at different cultivation times (0–28 h) and concentrations (2–2500 $\mu\text{g/mL}$)

bacterial growth of *L. paracasei* BCRC14023 at 18 h of fermentation (Fig. 7a) and showed its earlier effect on the growth of *L. rhamnosus* BCRC16000 after just 4 h of cultivation (Fig. 7b). These data pre-confirm that condensed tannin ELTB.3.1-2d possesses prebiotic effect.

For further determination of the potency of prebiotic activity, condensed tannin ELTB.3.1-2d was tested at a large concentration range of 2–2500 µg/mL, and cultivation time at 18 h and 24 h was chosen for growth detection of *L. paracasei* BCRC14023, while the growth of *L. rhamnosus* BCRC16000 was detected at 4 h and 24 h of fermentation. ELTB.3.1-2d showed its moderate enhancing effect on the growth of *L. paracasei* BCRC14023 with the relative bacterial growth of 114–141% in its concentration range of 5–625 µg/mL (detection at 18 h of cultivation time), while it demonstrated potential prebiotic activity with great relative activity of 158–216% in the concentration range of 2–1250 µg/mL (detection at 4 h of cultivation time) when *L. rhamnosus* BCRC16000 was used as probiotics. At longer time of cultivation (24 h), the growth enhancing effect of condensed tannin ELTB.3.1-2d on both bacterial strains was reduced due to the limitation of nutrients in the cultivation medium for the bacteria to survive [34]. Notably, the data also indicated that the condensed tannin ELTB.3.1-2d just showed its prebiotic activity in its concentration range of 5–625 µg/mL, and 2–1250 µg/mL when *L. paracasei* BCRC14023 and *L. rhamnosus* BCRC16000 were used as probiotics, respectively. At higher doses tested, this phytoconstituent may show its toxicity to these lactic bacterial strains. The prebiotic effect of vast natural products has been reported, including phenolic constituents [35–38]. However, this effect of condensed tannins has rarely been reported.

Conclusions

In this study, we investigated the parts used that were rich in condensed tannin and some major factors' effects on this substrate extraction. The results indicated that *E. laxiflorus* trunk bark is the richest source of condensed tannin among various parts used; condensed tannin was extracted from this part used with high scale by 75% methanol in water, at ≥ 50 °C. The condensed tannin ELTB.3.1-2d isolated from the *E. laxiflorus* trunk bark was qualified by NMR spectra and HPLC finger printing and evaluated as exhibiting vast beneficial activities. Condensed tannin ELTB.3.1-2d demonstrated potent activities of antioxidant, enzyme inhibition targets in type 2 diabetic treatment and prebiotic effect. Although the condensed tannin showed no anti-NO activity, it demonstrated moderate anticancer activity. Notably, condensed tannin ELTB.3.1-2d showed significant effect on reducing plasma glucose level in mice without diarrhea symptoms in sucrose tolerance test. The result suggested that *E. laxiflorus* is a rich source of natural bioactive products, and condensed tannin ELTB.3.1-2d extracted from this herbal is a potential natural product maybe a good candidate for health foods or drugs.

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Author contribution VBN and SLW took part in conceptualization and methodology; VBN involved in software analysis; VBN, SLW, TQT, TNN, ADN, and YHK involved in formal analysis; VBN, DNN, THN, NDH, MTN, CTD, TNT, and TTN involved in investigation; VBN took part in resource collection; VBN took part in data curation; VBN participated in original draft; SLW and VBN involved in review and editing; SLW and VBN took part in supervision; SLW and VBN participated in project administration.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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